



Fraud Detection Analysis & Modeling Report

1. Introduction

Fraud detection is a critical problem in financial systems due to the potential financial losses and impact on customer trust. The objective of this project is to analyze transaction data, identify patterns associated with fraudulent activity, and build a machine learning model capable of detecting fraud.

This project follows an end-to-end approach, covering data exploration, modeling, evaluation, and visualization using SQL, Python, and Power BI.

2. Dataset Overview

The dataset used is the **Credit Card Fraud Detection dataset**, containing anonymized transaction data.

Key features include:

- Time: Time elapsed between transactions
- Amount: Transaction value
- Class: Target variable (0 = Normal, 1 = Fraud)
- V1 – V28: PCA-transformed variables for privacy

The dataset contains **284,807 transactions**, of which only **492 are fraudulent**, making it highly imbalanced.

3. Data Exploration (SQL Analysis)

Initial analysis was performed using SQL to understand the dataset.

Key Findings:

- Fraud rate is approximately **0.17%**
- Fraud occurs more frequently during certain hours (peak around early morning)
- Most transactions (both fraud and normal) occur at **low amounts**
- Fraudulent behavior closely resembles normal transactions

Insight:

Fraud does not stand out clearly and often mimics legitimate behavior, making detection challenging.

4. Exploratory Data Analysis (Python)

Using Python, further analysis was performed:

- Confirmed the **high class imbalance**
- Observed a **right-skewed distribution** in transaction amounts
- Compared fraud vs normal transaction behavior using visualizations
- Identified weak individual correlations between features and fraud

Insight:

No single feature strongly predicts fraud — patterns are distributed across multiple variables.

5. Data Preparation

To prepare for modeling:

- Data was split into training and testing sets
 - Class imbalance was addressed using **oversampling of fraud cases**
 - Features were scaled using **StandardScaler** (for Logistic Regression)
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6. Model Development

Two models were tested:

6.1 Logistic Regression

- Captured fraud patterns effectively
- Achieved high recall
- Produced many false positives

6.2 Random Forest

- Achieved very high precision
 - Failed to detect most fraud cases
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7. Model Evaluation

Evaluation was performed using:

- Precision
- Recall
- Confusion Matrix

Results (Logistic Regression):

- Recall: ~**92%**
- Precision: ~**6%**

Interpretation:

- The model successfully identifies most fraudulent transactions
 - However, it incorrectly flags many legitimate transactions as fraud
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8. Business Insight

Fraud detection involves a trade-off:

- High Recall → catches fraud but increases false alarms
- High Precision → reduces false alarms but misses fraud

In this case, the model prioritizes **recall**, which is more suitable for fraud detection where missing fraud is costly.

9. Dashboard & Visualization

A Power BI dashboard was developed to present:

- Fraud rate and transaction overview

- Model performance (True/False Positives & Negatives)
- Precision and Recall metrics
- Fraud distribution insights

The dashboard highlights the imbalance in the dataset and the trade-offs in model performance.

10. Conclusion

This project demonstrates that:

- Fraud detection is complex due to class imbalance and subtle patterns
- Machine learning models can detect fraud but require careful evaluation
- No single metric is sufficient — trade-offs must be considered

The final solution provides both a predictive model and a visualization layer to support decision-making.

11. Future Improvements

- Improve precision using threshold tuning
- Experiment with advanced models (e.g., XGBoost)
- Incorporate real-world features (customer data, location, etc.)
- Deploy model for real-time fraud detection